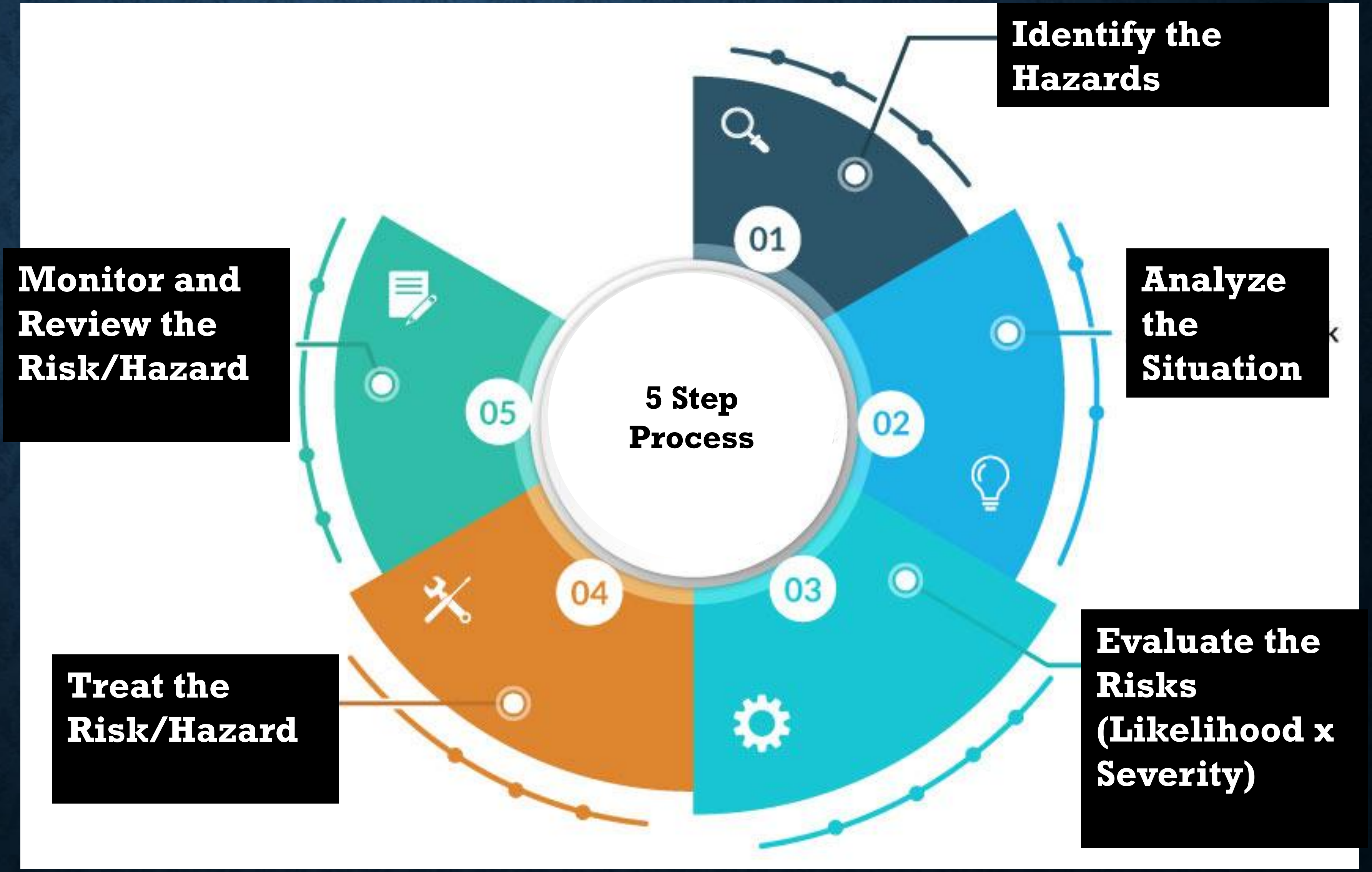
# 5-STEP PROCESS IS THE GLOBAL GOLD STANDARD FOR SAFETY MANAGEMENT.



# STEP 1: IDENTIFY THE HAZARDS

- The goal here is to find anything with the "potential to cause harm."
- You don't just look with your eyes; you look at manuals, past accident reports, and talk to workers.

**A large, heavy delivery truck reversing into a busy loading dock.**

- The Hazard: The moving vehicle (Kinetic energy/Mechanical hazard).
- Secondary Hazards: Blind spots for the driver, loud engine noise (masking shouts), and slippery floor surfaces in the dock.

# STEP 2: ANALYZE THE SITUATION

- Different people are at risk for different reasons.
- **The Driver:** Could hit a wall (whiplash).
- **Warehouse Staff:** Could be crushed between the truck and the dock.
- **Visitors/Pedestrians:** Might walk behind the truck, unaware of the blind spots.
- **How:** "Crushing injuries, bone fractures, or fatalities due to vehicle-pedestrian collision."

# EVALUATE THE RISKS (LIKELIHOOD X SEVERITY)

- We assign a numerical value to prioritize what needs fixing first.
- **Likelihood (1-3):** If trucks arrive 20 times a day and there are no barriers, the likelihood is **3 (High)**.
- **Severity (1-3):** A truck crushing a person is a **3 (Fatality/Catastrophic)**.
- **Risk Score:** $3 \times 3 = 9$

**3x3 Risk Matrix Template**

SEVERITY →

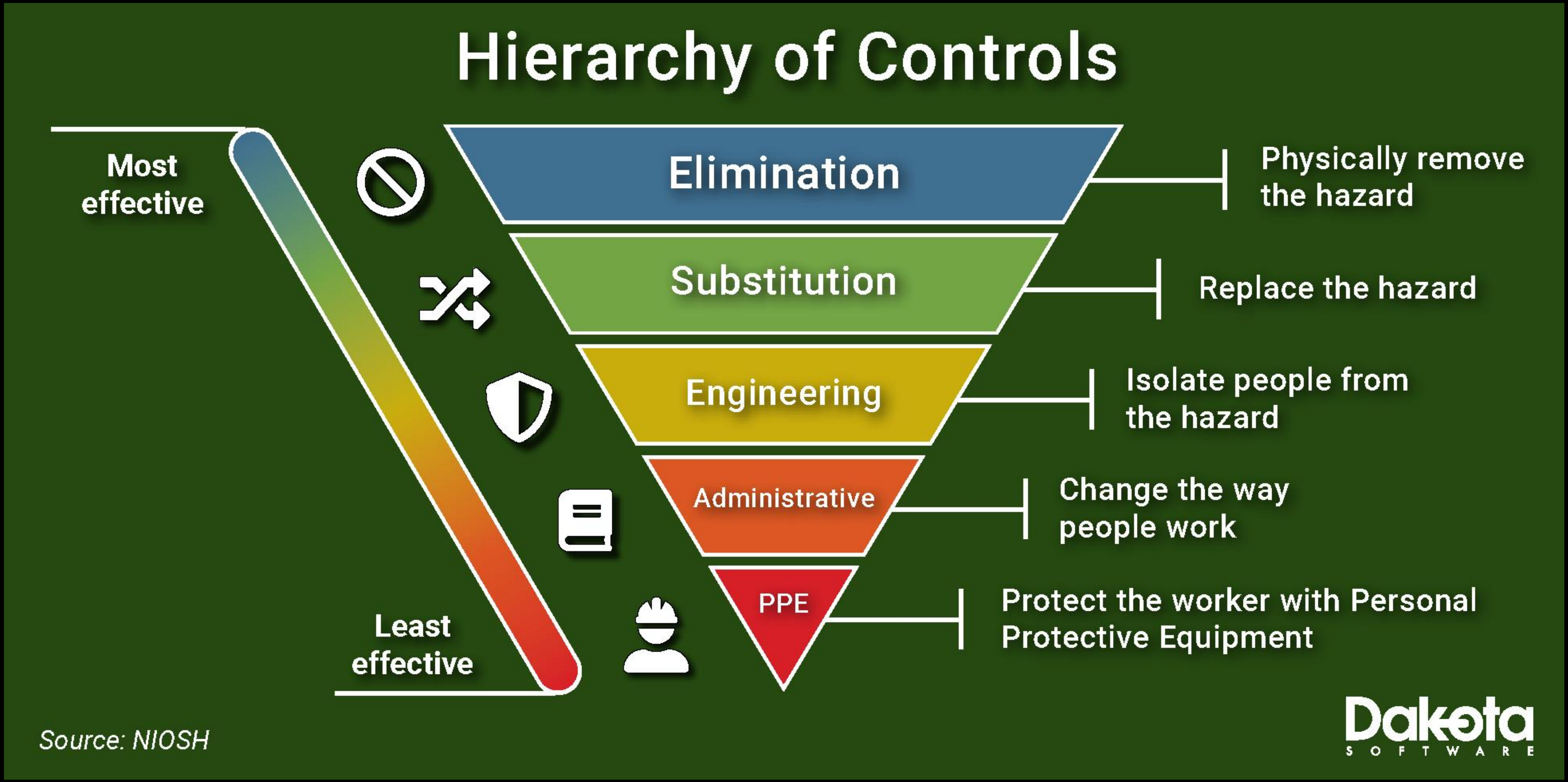
| LIKELIHOOD ↓ | 1 | 2 | 3 |
|---|---|---|---|
| 1 | LOW - 1 - | LOW - 2 - | MEDIUM - 3 - |
| 2 | LOW - 2 - | MEDIUM - 4 - | HIGH - 6 - |
| 3 | MEDIUM - 3 - | HIGH - 6 - | HIGH - 9 - |

# STEP 4: TREAT THE HAZARD OR RISK

The Hierarchy of Controls

Once a risk is identified, how do we fix it? We follow a strict order of effectiveness:

- Elimination: Physically remove the hazard.
- Substitution: Replace the hazard (e.g., use a less toxic chemical).
- Engineering Controls: Isolate people from the hazard (e.g., guardrails).
- Administrative Controls: Change the way people work (e.g., training, signs).
- PPE: Protect the worker with gear (e.g., gloves, helmets).



Hierarchy of Controls
Most effective
Least effective
Elimination
Substitution
Engineering
Administrative
PPE
Physically remove the hazard
Replace the hazard
Isolate people from the hazard
Change the way people work
Protect the worker with Personal Protective Equipment
Source: NIOSH
Dakota SOFTWARE

Step-by-Step Implementation Timeline

| Action | Responsible Party | Timeline |
| --- | --- | --- |
| Hazard assessment | Safety Officers | 1 Month |
| Implement engineering controls | Maintenance Team | 2-3 Months |
| Conduct training programs | HR & Safety Trainers | Ongoing |
| Provide PPE | Procurement Department | 1 Month |


# MONITORING AND REVIEWING CONTROLS

- Importance of regular monitoring to ensure controls remain effective.
- Methods - Inspections, audits, employee feedback, incident reports.
- Updating controls in response to changes in processes or regulations.

# SCENARIO 5: THE LABORATORY

- **Situation:** Researchers are using a highly volatile solvent to clean glass slides. The work is currently being done on a standard open workbench, and several staff members have complained of headaches and dizziness by the end of the day.

# SCENARIO 6: THE LOGISTICS WAREHOUSE

- Order pickers are required to lift 20kg boxes from floor-level pallets and place them onto a conveyor belt that is at waist height. This task is repeated roughly 150 times per shift. Two workers have reported chronic lower back pain.

# SCENARIO 5: THE LABORATORY

| Level of Control | Proposed Action |
|---|---|
| **Elimination** | Remove the solvent entirely and use a heat-cleaning process if applicable. |
| **Substitution** | Replace the volatile solvent with a water-based, non-toxic cleaning agent. |
| **Engineering** | Perform all slide cleaning inside a certified **Fume Hood** with active ventilation. |
| **Administrative** | Limit the time any one researcher spends on cleaning tasks (Job Rotation). |
| **PPE** | Provide high-grade respiratory masks and chemical-resistant gloves. |

# SCENARIO 6: THE LOGISTICS WAREHOUSE

| Level of Control | Proposed Action |
|---|---|
| **Elimination** | Redesign the workflow so that pallets arrive on elevated platforms, removing the need to lift from the floor. |
| **Substitution** | Use a vacuum lifting tool or a robotic arm to handle the weight of the boxes. |
| **Engineering** | Install a scissor-lift table that keeps the pallet at waist height as layers are removed. |
| **Administrative** | Mandatory "Safe Lifting" training and frequent stretching breaks throughout the shift. |
| **PPE** | Provide back support belts (though these are often considered the least effective). |